

Program :

```
def calculate_parity_bits(data):
    m = len(data)
    r = 0
    while (2 ** r) < (m + r + 1):
        r += 1
    return r

def generate_hamming_code(data):
    m = len(data)
    r = calculate_parity_bits(data)
    total_bits = m + r
    hamming = ['0'] * (total_bits + 1)
    j = 0
    for i in range(1, total_bits + 1):
        if i & (i - 1):
            hamming[i] = data[j]
            j += 1
    for p in range(r):
        position = 2 ** p
        parity = 0
        for i in range(1, total_bits + 1):
            if i & position:
                parity ^= int(hamming[i])
        hamming[position] = str(parity)
    return ''.join(hamming[1:])

def detect_and_correct(received):
    n = len(received)
    bits = ['0'] + list(received)
    error_position = 0
```

```

p = 1
while p <= n:
    parity = 0
    for i in range(1, n + 1):
        if i & p:
            parity ^= int(bits[i])
    if parity != 0:
        error_position += p
    p *= 2
if error_position == 0:
    print("No error detected.")
else:
    print("Error detected at position:", error_position)
    bits[error_position] = '1' if bits[error_position] == '0' else '0'
    print("Corrected codeword:", ''.join(bits[1:]))
data = ""
for i in range(1, n + 1):
    if i & (i - 1):
        data += bits[i]
print("Original data:", data)

```

```

data = input("Enter data bits: ")
codeword = generate_hamming_code(data)
print("\n--- TRANSMITTER SIDE ---")
print("Original data:", data)
print("Generated Hamming code:", codeword)
print("\n--- RECEIVER SIDE ---")
received = input("Enter received codeword: ")
detect_and_correct(received)

```

Outputs:

1) Without error :

```
Output Clear
Enter data bits: 1011

--- TRANSMITTER SIDE ---
Original data: 1011
Generated Hamming code: 0110011

--- RECEIVER SIDE ---
Enter received codeword: 0110011
No error detected.
Original data: 1011

=== Code Execution Successful ===
```

2) With error:

```
Output Clear
Enter data bits: 1011

--- TRANSMITTER SIDE ---
Original data: 1011
Generated Hamming code: 0110011

--- RECEIVER SIDE ---
Enter received codeword: 0110001
Error detected at position: 6
Corrected codeword: 0110011
Original data: 1011

=== Code Execution Successful ===
```